

Underwater Image Enhancement Based on Multichannel Adaptive Compensation

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Abstract—Due to underwater light absorption and scattering, underwater images often suffer from color distortion and low contrast. However, existing underwater image enhancement methods can only solve the problem of underwater image degradation in a specific scene and do not consider the impact of dynamic changes in water depth and water quality on underwater image degradation. To solve the above problems, this article proposes a multichannel adaptive fusion underwater image enhancement algorithm. First, to alleviate the color distortion issue caused by light attenuation, this article proposes a gridded adaptive channel compensation algorithm. Subsequently, the compensated image is used for multichannel image enhancement. The first channel image is the enhanced result of the compensated image using the local entropy-constrained gray world algorithm proposed in this article and the second channel image is the enhanced result of the first channel image using the contrast limited adaptive histogram equalization (CLAHE) algorithm. Next, the saliency weight image and brightness weight image of the two-channel images are calculated respectively. Finally, to combine the information advantages of different channel images, this article adopts a Laplacian–Gaussian pyramid to fuse the two-channel enhanced images and their corresponding weight images to obtain the final enhanced image. Experiments on three datasets, LSUI, UIEB, and RUIE, show that the underwater images processed by the algorithm proposed in this article have a good improvement in color and contrast, and robustness is better than the comparative methods. The underwater image quality measure metric (UIQM) value of the enhanced images is 126.844% higher than the original images, the underwater color image quality evaluation metric (UCIQE) value is 66.370% higher than the original images, and the number of ORB feature points is 95.862% higher than the original images. Results are available at <https://drive.google.com/file>.

Index Terms—Channel compensation, entropy, gray world, multichannel adaptive fusion, underwater image enhancement.

I. INTRODUCTION

THE ocean contains abundant resources, and the efficient development and utilization of marine resources are crucial for addressing the current world energy crisis. Underwater images serve as important carriers for understanding the distribution of various resources in the ocean. Therefore, the quality of images is particularly important for marine resource

exploration [1], [2], [3]. However, underwater imaging is more complex than air imaging, as shown in Fig. 1, different light wavelengths have different attenuation rates underwater. The longest wavelength, red light, experiences the highest level of attenuation in water, while the attenuation rates for blue and green light are relatively lower, which causes blue and green color distortion in underwater images [4], [5]. At the same time, suspended particles in the water can also affect the propagation of light, leading to the blurring of image details [4]. In addition, when the camera reaches a certain depth, it can only rely on artificial light sources for imaging, but artificial light sources can cause the occurrence of light spots, resulting in shadows in the captured images [6], [7]. The above problems will lead to color distortion and low contrast in underwater images, which cannot provide high-quality images for marine resource exploration.

To obtain underwater images with higher visual quality, underwater image enhancement methods based on physical models, nonphysical models, and deep learning have been well explored [8]. Traditional nonphysical model methods cannot simultaneously solve the problem of image quality degradation caused by light selective attenuation and uneven illumination. The traditional physical model methods establish a mathematical model based on the underwater image degradation principle and then obtain enhancement images by inversely solving the parameters of the underwater imaging degradation model. Water temperature, chemical fertilizers, and other factors can influence the growth of algae and microorganisms in water [9], but algae and microorganisms in water directly affect the water quality. This causes the transmittance of the light source underwater to be dynamically changing, making it difficult to enhance underwater images through the construction of underwater imaging physical models. The deep-learning methods rely on a large amount of underwater images for modeling and training, and the amount of images directly influences the efficacy of image enhancement. However, the underwater reference images are difficult to collect, and most of the reference images of existing underwater image datasets are synthetic.

Based on the above problems, this article proposes a multichannel adaptive fusion underwater image enhancement algorithm, which can dynamically solve the color distortion and low contrast problems caused by light attenuation in different scenes. First, to alleviate the color distortion problem caused by light attenuation, this article proposes a grid adaptive channel compensation algorithm. Second, based on the compensated image, multichannel image enhancement

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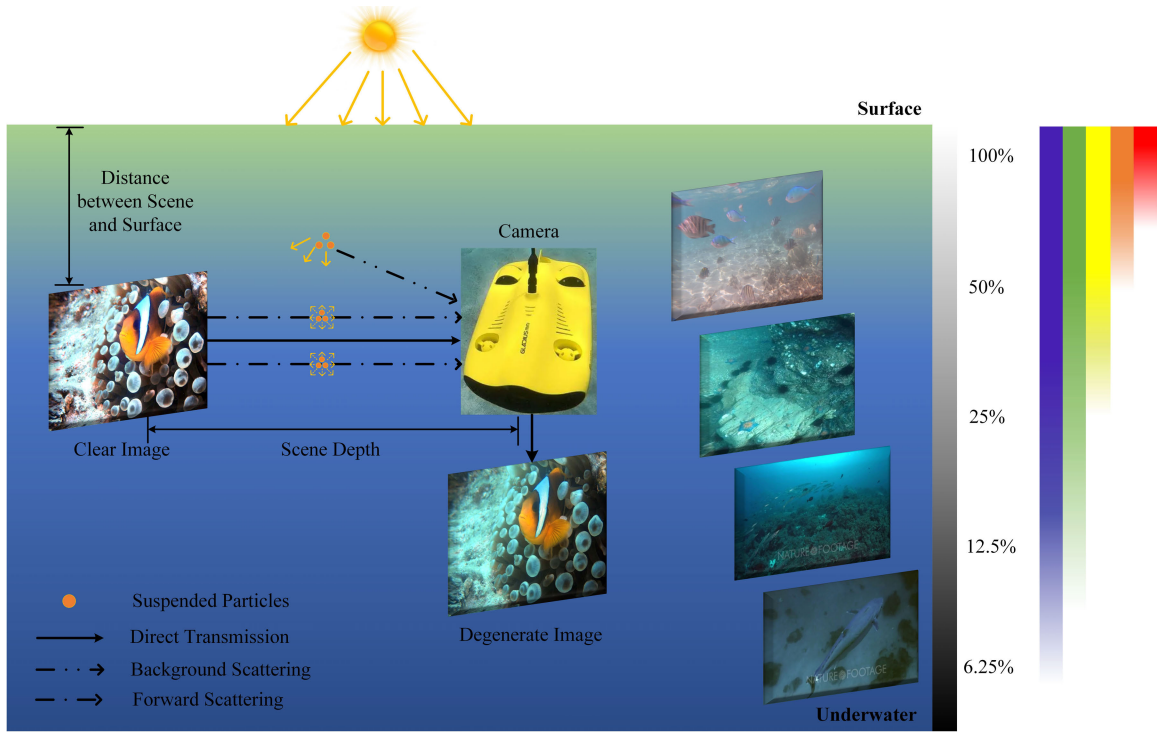


Fig. 1. Underwater optical imaging model.

fusion is carried out. To further solve the problem of underwater image color distortion, this article proposes a local entropy-constrained gray world algorithm, and the enhanced result is used as the first channel image. At the same time, to solve the problem of low image contrast, the contrast-limited adaptive histogram equalization (CLAHE) algorithm is used to enhance the contrast of the first channel image, and the enhanced result is used as the second channel image. Third, the saliency weight map and brightness weight map of the two-channel images are calculated respectively. Finally, to combine the information advantages contained in different channel images and prevent the occurrence of artifacts, this article uses the Laplacian–Gaussian pyramid to fuse the two-channel enhanced image and the corresponding weight maps. The contributions of the article are summarized as follows.

- 1) This article proposes a multichannel adaptive fusion underwater image enhancement method, which can effectively solve the problems of color distortion and low contrast caused by light attenuation in different scenes.
- 2) To solve the problem of selective attenuation of underwater light, this article proposes a gridded adaptive channel compensation algorithm.
- 3) To solve the color distortion problem of underwater images, this article proposes a local entropy-constrained gray world white balance algorithm.

The rest of this article is organized as follows. In Section II, we review the related work on underwater image enhancement. In Section III, we provide a detailed introduction to the proposed multichannel adaptive fusion underwater image enhancement algorithm. In Section IV, we describe the selected datasets, experimental setup, and detailed

experimental results. In Section V, we provide a concluding summary and indicate future research directions.

II. RELATED WORK

In the field of underwater image enhancement, scholars have conducted extensive research. Currently, underwater image enhancement methods can be classified into three categories: physical models [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], nonphysical models [22], [23], [24], and deep learning [25], [26], [27], [28], [29], [30], [31], [32]. Methods based on physical models for underwater image enhancement heavily rely on the underwater optical imaging model and obtain clear underwater images by inversely solving the parameters of the underwater imaging model. However, this type of method is easily affected by the suspended particles in the scene and artificial light sources during the process of parameter estimation [33], [34], [35]. Most of the existing underwater image enhancement methods based on physical models are developed based on the Jaffe–McGlamery physical model [36], [37]. The imaging process of the Jaffe–McGlamery underwater optical model is shown in Fig. 1. This model represents the imaging of underwater images as a linear combination of direct component, forward-scattering component, and backscattering component. The Jaffe–McGlamery model is primarily influenced by two parameters of background light and transmission coefficient [22]. Different scholars have investigated diverse methods for solving these two parameters to obtain clear underwater images. Drews et al. [11] proposed an underwater image enhancement algorithm based on statistical priors by analyzing the characteristics of underwater imaging, but this method is less robust and is easily affected by scene

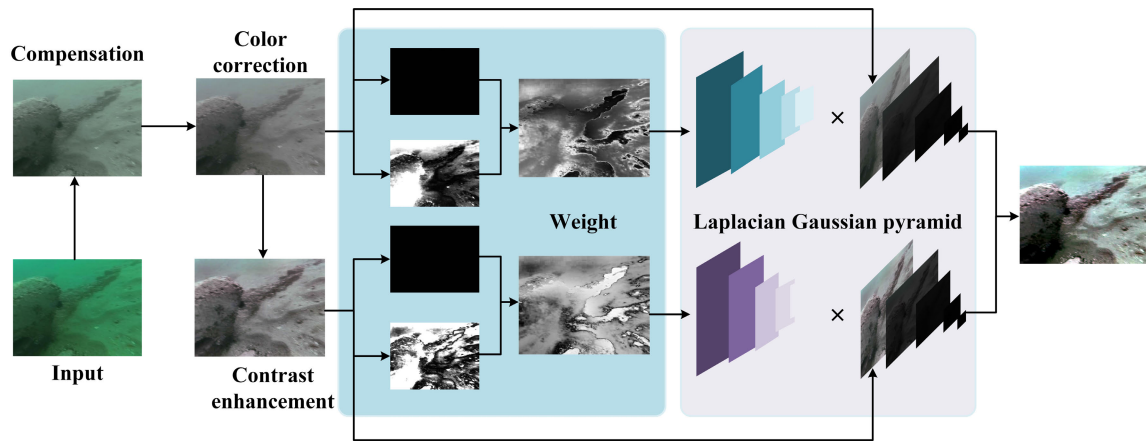


Fig. 2. Overall flowchart of the underwater image enhancement based on multichannel adaptive compensation.

characteristics. Dai et al. [14] proposed a hybrid underwater image enhancement method that effectively estimates the background light and transmittance from a single image to achieve underwater image enhancement. Yu et al. [16] proposed an underwater image enhancement algorithm based on the color line model and homomorphic filtering to solve the problems of low contrast and low visibility of underwater images. Ding et al. [17] proposed a scene depth regularization underwater image enhancement method, which refines the imaging model through variational regularization to achieve smooth details while retaining more texture information. Yang et al. [18] proposed a low-complexity underwater image enhancement method based on the dark channel prior, which can achieve real-time requirements while enhancing underwater images. Wen et al. [19] proposed an underwater imaging optical model by studying the difference of light attenuation in the atmosphere and underwater, which can effectively enhance the images captured in turbid waters. Song et al. [21] proposed a depth estimation model based on underwater light attenuation prior, which can achieve underwater image enhancement in a short time.

Methods based on nonphysical models for underwater image enhancement mainly utilize digital image-processing techniques to enhance specific metrics (such as contrast, saturation, etc.) to improve the visual quality of the underwater image [38]. Ancuti et al. [22] adopted a multiscale fusion strategy to fuse two channel weight maps obtained by gamma correction and sharpening methods, which can effectively solve the problem of image overexposure and achieves significant enhancements in image contrast and edges. Zhuang et al. [23] proposed a Bayesian retina algorithm with multiple-step priors to solve the problems of color distortion and low contrast. Zhang et al. [24] proposed an attenuation channel compensation matrix to correct the color of underwater images based on the phenomenon of differential attenuation in different color channels.

Methods based on deep learning for underwater image enhancement mainly estimate the underwater imaging model parameters using a large amount of underwater image data to achieve image enhancement [39], [40], which requires a large number of underwater images for training. However, the existing underwater image datasets [30], [41], [42], [43],

[44], [45] have disadvantages such as few scenes, insufficient data, and difficulty in obtaining reference images without degradation. Hou et al. [26] proposed a deep-learning model URCNN, which utilizes joint residual learning to estimate underwater illumination for haze removal and color correction enhancement of underwater images. Cao et al. [27] proposed a combination network to solve the issues of color distortion and low contrast of underwater images, which is divided into two branches to realize the estimation of background light and scene depth, respectively. Islam et al. [30] proposed the underwater image enhancement network model FUnIE-GAN to solve the problem of low brightness and color distortion, which can meet real-time requirements while ensuring image quality. Yang et al. [31] proposed a conditional generative adversarial network cGAN to achieve underwater image enhancement. Huang et al. [32] introduced a novel adaptive group attention module and combined it with the swin transformer to achieve enhancement of underwater images. Li et al. [45] aiming at the problem of lack of real underwater images, created an underwater image enhancement benchmark, and proposed a network model to solve the problem of underwater image quality degradation. Saleh et al. [46] proposed a novel unsupervised underwater image enhancement method to solve the problem of color distortion and low contrast of underwater images, which uses a conditional variational autoencoder to implement the training of the network model. Naik et al. [47] proposed a shallow neural network that can enhance underwater images in real time.

III. METHOD

This section will offer a comprehensive introduction to the framework of underwater image enhancement based on multichannel adaptive compensation. First, the adaptive channel compensation mechanism for the complex degradation of underwater images is introduced. Subsequently, the underwater image color correction and contrast enhancement mechanism is introduced. Finally, the multichannel image fusion enhancement mechanism is introduced. Fig. 2 shows the algorithm framework of underwater image enhancement based on multichannel adaptive compensation proposed in this article.

Algorithm 1 Gridded Adaptive Channel Compensation Algorithm

Input: degraded image: I_{in} , the length and width of the degraded image: L, W , the length and width of the gridded sub-image: l, w , gain coefficient: ω

Output: Color compensation sub-images: I_{sub_comp} , Number of sub-images: N .

```

1 if  $L\%l = 0$  then
2   Grid the degraded image
3 else
4   Grid the degraded image according to the method
   shown in Fig. 3.
5 Label the sub-images( $1 \cdots N(\frac{L}{l} \times \frac{W}{w})$ )
6 if  $i < N$  then
7   Calculate the mean value  $\bar{R}, \bar{G}, \bar{B}$  of the  $R, G$ , and
    $B$  channel of the sub-image.
8   Use formula 1 to calculate the maximum value
    $V_{max}$ .

$$V_{max} = \max(\bar{R}, \bar{B}, \bar{G}). \quad (1)$$

9   Find the minimum attenuation channel  $V_i$ 
   corresponding to  $V_{max}$ .
10  Use formula 2 to compensate for the severely
   attenuated channel.

$$\begin{cases} G = G + \omega \times (V_{max} - \bar{G}) \times (1 - \bar{G}) \times V_i \\ R = R + \omega \times (V_{max} - \bar{R}) \times (1 - \bar{R}) \times V_i \\ B = B + \omega \times (V_{max} - \bar{B}) \times (1 - \bar{B}) \times V_i. \end{cases} \quad (2)$$

11  Use formula 3 to limit the pixel value of the
   compensated image.
12  for  $j \leftarrow R, G, B$  do
13    
$$V_i = \begin{cases} 0, & V_i < 0 \\ V_i, & \text{else} \\ 255, & V_i > 255 \end{cases} \quad (3)$$

14 return  $I_{sub\_comp}, N$ 

```

A. Gridded Adaptive Channel Compensation

The quantity and size of suspended particles in water will influence light scattering, and different colors of light have different attenuation rates in water, so the color distortion in images taken at different water depths is also different. When there are a lot of suspended particles, the camera imaging tends to appear green. When capturing images in the deep sea, the camera imaging tends to appear blue. The problem of blue or green color distortion directly influences the quality of the image, which makes underwater images unsuitable for deep-sea exploration. Based on this problem, this article proposes a gridded adaptive channel compensation algorithm, which can adaptively compensate the attenuated channels based on the attenuation status of the R, G , and

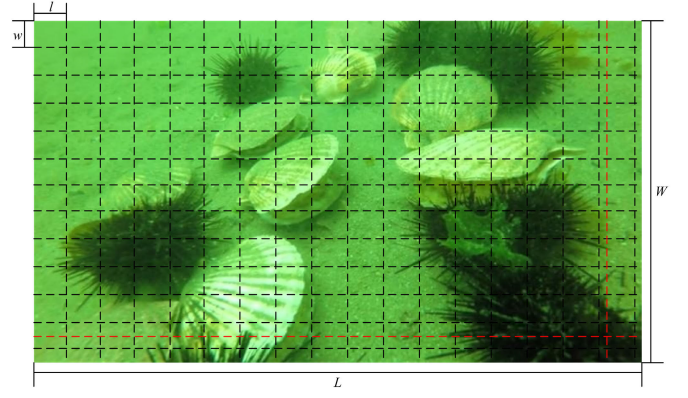


Fig. 3. Gridded image.

B channels, thereby avoiding overcompensation. Algorithm 1 summarizes the implementation steps of the gridded adaptive channel compensation algorithm.

B. Underwater Image Color Correction

1) *Gray World White Balance*: When using a camera to capture an image, the camera output image expression is as follows:

$$I_c(x) = \int_l [c(\lambda)E(\lambda)S(x, \lambda)]d\lambda \quad (4)$$

where c represents the pixel value of the R, G , and B channel image; l represents the wavelength range of the visible spectrum; x represents the spatial coordinate; λ represents the wavelength of the light; $c(\lambda)$ represents the photosensitive function of the imaging device; $E(\lambda)$ represents the spectral power distribution of the light source; and $S(x, \lambda)$ represents the spectral reflectance of the surface of the object.

Assuming that the image is imaged under a single light source, the purpose of color constancy is to estimate the light source

$$e = \begin{pmatrix} R_e \\ G_e \\ B_e \end{pmatrix} = \int_l c(\lambda)E(\lambda)d\lambda. \quad (5)$$

Gray world assumption suggests that in the objective world, the color variations of objects and their surroundings are random, independent, and irrelevant. For an image with rich color variations, the average values of the red, green, and blue channel images are approximately equal, that is, the average reflectance coefficient in the scene is a gray value β , which can be expressed as follows:

$$\frac{\int_X S(x, \lambda)dx}{\int_X dx} = \beta \quad (6)$$

where X represents the image pixel coordinate.

From the combination of formulae (4)–(6), it can be inferred that the light source color can be obtained by calculating the mean values of the R, G , and B of the captured image, as shown in the following equation:

$$\frac{\int I_c(x)dx}{\int dx} = \frac{\int \int_l [c(\lambda)E(\lambda)S(x, \lambda)]d\lambda dx}{\int dx}$$

$$\begin{aligned}
 &= \int_l E(\lambda) \left(\frac{S(x, \lambda) dx}{\int dx} \right) c(\lambda) d\lambda \\
 &= \beta \int_l E(\lambda) c(\lambda) d\lambda \\
 &= \beta e.
 \end{aligned} \tag{7}$$

Formulae (4)–(7) are the mathematical derivation process of the gray world assumption. Based on this assumption, Buchsbaum [48] proposed the gray world white balance algorithm. The specific calculation steps of the gray world algorithm are as follows.

- 1) Calculate the average values $\bar{R}$, $\bar{G}$, $\bar{B}$ of the R, G, and B channels of the captured image, respectively.
- 2) Calculate the average reflectance coefficient β of the captured image. Generally, there are two methods to calculate the coefficient β .
 - a) The first method is to directly set it as a fixed value.
 - b) The second method is to take the average of the mean values of the image in each channel of the captured image, as shown in formula (8). In this article, the second method is used to calculate the average reflectance coefficient β

$$\beta = \frac{\bar{R} + \bar{G} + \bar{B}}{3}. \tag{8}$$

- 3) Use formula (9) to calculate the gain coefficients of the R, G, and B channels

$$\begin{cases} \beta_r = \frac{\beta}{\bar{R}} \\ \beta_g = \frac{\beta}{\bar{G}} \\ \beta_b = \frac{\beta}{\bar{B}} \end{cases} \tag{9}$$

- 4) Based on the VonKries model [49], use the gain coefficient to enhance each channel of the captured image, as shown in the following equation:

$$I_{\text{new}} = \begin{bmatrix} \beta_r & & \\ & \beta_g & \\ & & \beta_b \end{bmatrix} I_{\text{old}} \tag{10}$$

where I_{new} represents the captured image and I_{old} represents the enhanced image.

2) Local Entropy-Constrained Gray World Algorithm:

According to the gray world assumption, the gray world algorithm works well when the captured image has rich colors, but it performs poorly when the captured image has few colors. To solve this problem, this article proposes a local entropy-constrained gray world algorithm. Image entropy is a statistical form of feature, which represents the aggregation feature of the gray distribution in an image, as shown in the following equation:

$$H = - \sum_{i=0}^{255} p_i \log p_i \tag{11}$$

where H represents the image entropy value; i represents the gray value; and $p(i)$ represents the probability of gray value i occurs in the gray image.

To calculate the entropy of a color image, it is necessary to first compute the image entropy of the R, G, and B channels separately and then calculate their average as the image entropy. The local entropy of the image reflects the intensity of the gray level variation in the local area of the image. When the color of the area is richer, the entropy value of the area is greater. The entropy value of an image is not only related to the color richness, but also related to the grayscale representation. The gray values of R , G , and B single-channel images are 256 levels, which requires 8 bits to represent 256 levels of grayscale. Therefore, the minimum value of H is 0, and the maximum value is 8.

To solve the shortcomings of the gray world algorithm, which leads to large color correction errors due to fewer colors, this article uses local entropy to judge the local color richness of the image. If the local entropy is large, it indicates that the region has a high color richness and can perform color correction using the gray world algorithm, otherwise, the local area will not be corrected. Algorithm 2 summarizes the implementation steps of the local entropy-constrained gray world algorithm.

Algorithm 2 Local Entropy-Constrained Gray World Algorithm

Input: degraded image: I_{in}

Output: Color correction result: $I_{\text{corr_out}}$

```

1  $I_{\text{sub\_comp}}, N \leftarrow \text{Algorithm 1}(I_{\text{in}})$ 
2 if  $i < N$  then
3   calculate the average values  $\bar{R}$ ,  $\bar{G}$ ,  $\bar{B}$  of the R, G,
   and B channels of the local subimage,
   respectively.
4   calculate the local subimage average reflectance
   coefficient  $\beta$  based on formula 8.
5   calculate the gain coefficients  $[\beta_r, \beta_g, \beta_b]$  of the
   three channels of  $R$ ,  $G$ , and  $B$  of the subimage
   based on formula 9.
6   calculate the entropy value  $[H_r, H_g, H_b]$  of the
   subimage based on formula 11.
7   use the formula 12 to constrain the gain coefficient
    $[\beta_r, \beta_g, \beta_b]$  of the subimage based on the entropy
    $[H_r, H_g, H_b]$ .
8   for  $n \leftarrow R, G, B$  do
9     
$$\beta_n^f = \left( \frac{\beta_n}{V} - 1 \right) \times \frac{H_n}{8} + 1 \tag{12}$$

10    substitute the gain coefficient  $[\beta_r^n, \beta_g^n, \beta_b^n]$  into
    formula 10 to correct the sub-image.
11 return  $I_{\text{corr\_out}}$ 
    
```

C. Underwater Image Contrast Enhancement

Due to the severe attenuation of underwater light, the captured images often suffer from low contrast. Histogram equalization [50], [51] is a commonly used grayscale transformation method that can transform the histogram of the

captured image into a uniform distribution, thereby enhancing the contrast of the image. Therefore, this article adopts the CLAHE algorithm [52] to improve the contrast of the captured image.

D. Weight Image Calculation

To further enhance the quality of underwater images, this article fuses the color-corrected image and the contrast-enhanced image using a special weight image. The attenuation of light results in poor brightness of underwater images, and the suspended particles can lead to blurred details of underwater images. In view of the above two cases, this article uses the saliency weight image and the brightness weight image to fuse the images.

1) *Saliency Weight*: To distinguish objects from the background in an underwater image, highlight the salient area, and enhance the contrast between the target area and the background area, this article first performs Gaussian filtering on the enhanced image to remove the inconspicuous areas in the image and then converts the image from the RGB color space to LAB color space. Subsequently, the saliency weight of the enhanced image is computed using as follows:

$$W_{\text{sig}} = (L - \bar{L})^2 + (A - \bar{A})^2 + (B - \bar{B})^2 \quad (13)$$

where $\bar{L}$, $\bar{A}$, and $\bar{B}$ represent the average values of channel L , channel A , and channel B , respectively.

2) *Brightness Weight*: To make the enhanced image have better-detailed texture, this article introduces a brightness weight mechanism to assign higher weights to pixels with favorable brightness. First, use formula (14) to grayscale the image. Subsequently, normalize the grayscale image. Finally, use formula (15) to calculate the brightness weight of the image

$$\begin{aligned} \text{gray}(x, y) &= 0.299 \times R(x, y) + 0.587 \times G(x, y) \\ &\quad + 0.114 \times B(x, y) \end{aligned} \quad (14)$$

where $R(x, y)$, $G(x, y)$, and $B(x, y)$ represent the red, green, and blue components of an RGB image, respectively,

$$S_{\text{bri}}(x, y) = \exp\left(-\frac{(\text{gray}(x, y) - 0.5)}{2\sigma^2}\right) \quad (15)$$

where σ represents the standard deviation, whose default value is 0.25.

3) *Normalized Weight*: The saliency weight image and brightness weight image of color correction image and contrast enhancement image are calculated using formulae (13) and (15). Then, the saliency weight and brightness weight are normalized using the following equation:

$$\begin{cases} W_1 = \frac{W_{\text{sal1}} + S_{\text{bri1}}}{W_{\text{sal1}} + S_{\text{bri1}} + W_{\text{sal2}} + S_{\text{bri2}}} \\ W_2 = \frac{W_{\text{sig2}} + S_{\text{bri2}}}{W_{\text{sal1}} + S_{\text{bri1}} + W_{\text{sal2}} + S_{\text{bri2}}} \end{cases} \quad (16)$$

where W_1 is the normalized weight of the color correction image. W_2 is the normalized weight of the contrast enhancement image.

E. Laplacian–Gaussian Pyramid Fusion

To combine the advantageous information from different source images to generate a high-quality underwater image, based on the idea of an image Laplacian–Gaussian pyramid, this article uses the normalized weights obtained in Section III-D to fuse the color correction images obtained in Section III-B and the contrast enhancement images obtained in Section III-C. The Laplacian–Gaussian pyramid is a multiscale fusion method, which can reduce the uncertainty and error that may exist in the background and maximize the use of image advantage information provided by various information sources. The fusion steps of the Laplace–Gaussian pyramid are as follows.

- 1) First, separate the input image into R, G, and B single-channel images, which are used as the G_0 layer of the Gaussian pyramid.
- 2) Then, Gaussian convolution filtering and downsampling are performed on the G_0 layer of the Gaussian pyramid to obtain the G_1 layer of the Gaussian pyramid. The G_1 layer of the Gaussian pyramid is processed in the same way to obtain the G_2 layer of the Gaussian pyramid. By analogy, obtain the image of each layer of the Gaussian pyramid. The calculation formula is shown as follows:

$$G_N(x, y) = \sum_{i=-2}^2 \sum_{j=-2}^2 w(i, j) G_{N-1}(2x + i, 2y + j) \quad (17)$$

where N represents the number of Gaussian pyramid layers, and $N = 5$ in this article. $w(i, j)$ represents 5×5 Gaussian convolution kernel.

- 3) Based on the Gaussian pyramids of R, G, and B channel images obtained in step 2), calculate the corresponding Laplacian pyramids using as follows:

$$\text{LP}_N = \begin{cases} G_N(x, y) - 4 \sum_{i=-2}^2 \sum_{j=-2}^2 w(i, j) \times G_{N+1}\left(\frac{x+i}{2}, \frac{y+j}{2}\right) & 0 \leq N < 5 \\ G_N(x, y) & N = 5. \end{cases} \quad (18)$$

- 4) Calculate the Gaussian pyramids of the significance weight image and the brightness weight image using (17).
- 5) Fuse the Laplacian pyramid obtained in step 3) and the Gaussian pyramid obtained in step 4) at various scales. The calculation formula is shown as follows:

$$F_N(x, y) = \sum_{m=1}^2 [\text{LP}_N]_m(x, y) [G_N]_m(x, y) \quad (19)$$

where m represents the number of enhanced images; N represents the number of layers of the pyramid, the range is from 1 to 5; $F_N(x, y)$ represents the image pyramid after fusion; $[\text{LP}_N]_m(x, y)$ represents the N th layer Laplacian image derived from processing the m th enhanced image through step 3); $[G_N]_m(x, y)$ represents the N th layer Gaussian pyramid image derived from processing the m th weighted image through step 4).

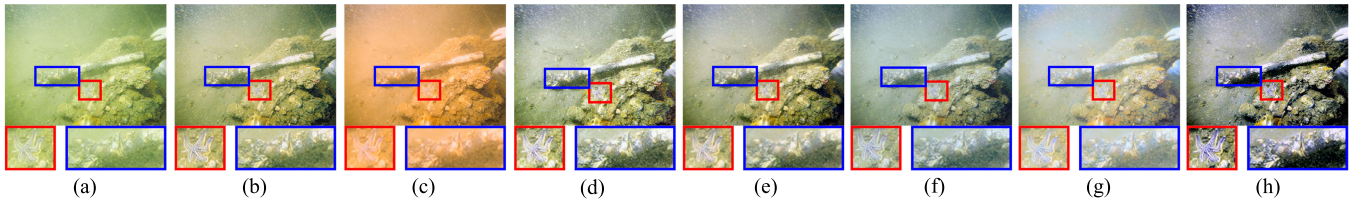


Fig. 4. Visual results on the UIEB dataset. (a) Original image. (b) Li et al. [45]. (c) Naik et al. [47]. (d) Wang et al. [53]. (e) Liu et al. [54]. (f) Saleh et al. [46]. (g) Rao et al. [55]. (h) Ours.

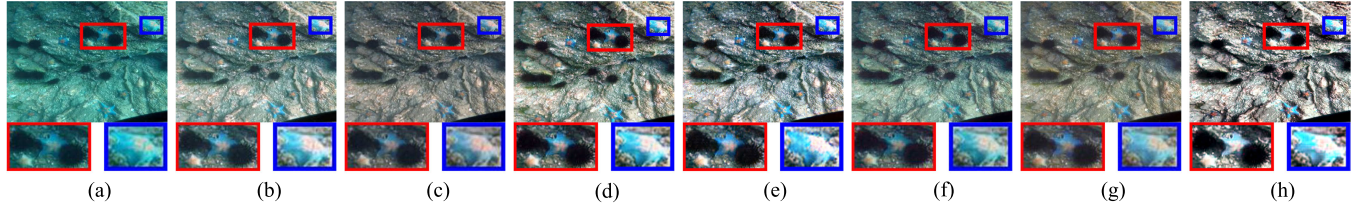


Fig. 5. Visual results on the RUIE dataset. (a) Original image. (b) Li et al. [45]. (c) Naik et al. [47]. (d) Wang et al. [53]. (e) Liu et al. [54]. (f) Saleh et al. [46]. (g) Rao et al. [55]. (h) Ours.

- 6) Carry out an upsampling operation on the image pyramid obtained in step 5) to obtain the final enhanced image.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Datasets

To verify the robustness of the algorithm proposed in this article, we conducted experimental verification and analysis on the public datasets LSUI [41], RUIE [42], and UIEB [45].

1) *UIEB*: The UIEB dataset contains 950 underwater images, which are divided into two subsets. One subset contains 890 original underwater images and corresponding reference images, and the other subset contains 60 underwater degraded images. We randomly selected 50 images from the UIEB dataset as test images.

2) *RUIE*: The RUIE dataset contains 4230 underwater images, which are divided into three subsets: Underwater Image Quality Set (UIQS), Underwater Color Cast Set (UCCS), and Underwater Higher-level Task-driven Set (UHTS). The UHTS subset includes 300 images containing a variety of marine organisms, which can be used to evaluate the impact of underwater image enhancement algorithms on high-level computer tasks. We randomly selected 125 images from the RUIE dataset as test images.

3) *LSUI*: The LSUI dataset contains 4279 underwater images. We randomly selected 125 images from the LSUI dataset as test images.

B. Comparative Experiments of the Multichannel Adaptive Fusion

1) *Qualitative Results*: To verify the effectiveness and robustness of the algorithm proposed in this article, we compared it with typical underwater image enhancement algorithms on LSUI, RUIE, and UIEB datasets. The typical underwater image enhancement algorithms selected in this article include: Li et al. [45], Naik et al. [47], Wang et al. [53], Liu et al. [54], Saleh et al. [46], and Rao et al. [55]. The visual comparisons of the algorithm proposed in this article and the comparison algorithm are shown in Figs. 4–6.

It can be seen from the comparison of Figs. 4–6 that the algorithm proposed by Li et al. [45] can solve the color distortion problem to a certain extent, but the processed image still has color distortion and low contrast. The algorithm proposed by Naik et al. [47] can effectively solve the color distortion problem of LSUI and RUIE datasets, but it cannot solve the color distortion problem shown in the UIEB dataset and has poor robustness. The algorithms proposed by Wang et al. [53], Liu et al. [54], Saleh et al. [46], and Rao et al. [55] can alleviate the color distortion problem, but the contrast is low and the details are blurred in the enhanced images. In contrast, the algorithm proposed in this article can effectively solve the problems of color distortion and low contrast, and the details in the enhanced image are also clearer.

2) *Quantitative Results*: To objectively evaluate the enhancement results of each method, this article selects the commonly underwater no-reference evaluation metrics UCIQE [56] and UIQM [57] to quantify the enhancement results. The purpose of underwater image enhancement is to realize the navigation, localization, and tracking of underwater robots [58], the extraction of feature points is very important for these functionalities. Therefore, this article uses the number of extracted ORB feature points [59] to quantitatively evaluate the enhancement results of various methods. The underwater color image quality evaluation metric (UCIQE) comprehensively evaluates the image quality from the hue, saturation, and contrast. The underwater image quality measure metric (UIQM) comprehensively evaluates the image from the color saturation, sharpness, and contrast, which is proposed based on the human visual perception system. Table I shows the UCIQE, UIQM, and ORB feature point extraction values corresponding to the enhancement results of different algorithms on UIEB, RUIE, and LSUI datasets. It can be seen from Table I that the image processed by the algorithm proposed in this article has achieved optimal results in terms of UIQM, UCIQE, and the number of ORB feature point extraction. The average value of UIQM is increased by 126.844% compared with the original image, the average value of UCIQE is increased by 66.370% compared with the original image, and the average number of

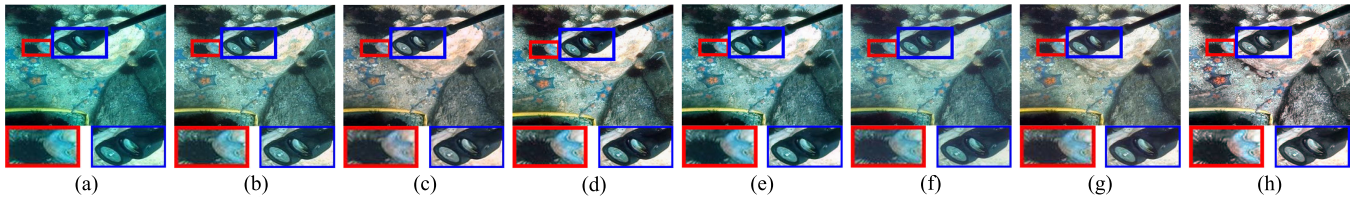


Fig. 6. Visual results on the LSUI dataset. (a) Original image. (b) Li et al. [45]. (c) Naik et al. [47]. (d) Wang et al. [53]. (e) Liu et al. [54]. (f) Saleh et al. [46]. (g) Rao et al. [55]. (h) Ours.

TABLE I

QUANTITATIVE EVALUATION ON THE UIEB [45], RUIE [42], AND LSUI [41] DATASETS, USING UIQM [57], UCIQE [56], AND NOFP (NUMBER OF ORB FEATURE POINTS) [59] METRICS. THE LARGER THE METRIC, THE BETTER

Dataset	Metrics	Methods							
		Orig.	Li et al [45]	Naik et al [47]	Wang et al [53]	Liu et al [54]	Saleh et al [46]	Rao et al [55]	Ours
UIEB	UIQM	2.246	3.838	3.071	4.189	3.876	2.574	4.209	4.870
	UCIQE	0.429	0.586	0.426	0.652	0.652	0.512	0.434	0.692
	NOFP	13541	19219	13352	23718	20444	14727	13525	27623
RUIE	UIQM	2.009	4.396	3.553	4.705	4.549	3.358	4.479	5.223
	UCIQE	0.440	0.595	0.431	0.630	0.699	0.539	0.444	0.771
	NOFP	2356	4295	2290	4617	5804	3011	2374	8039
LSUI	UIQM	2.496	3.921	3.344	4.825	4.418	3.180	4.616	5.218
	UCIQE	0.478	0.599	0.466	0.649	0.680	0.556	0.483	0.776
	NOFP	14700	19398	13708	20395	20506	17662	14966	24267
Average	UIQM	2.250	4.052	3.323	4.573	4.281	3.037	4.435	5.104
	UCIQE	0.449	0.593	0.441	0.644	0.677	0.536	0.454	0.747
	NOFP	10199	14304	9783	16243	15584	11800	10288	19976

ORB feature points is increased by 95.862% compared with the original images.

C. Underwater Object Detection

1) *Experimental Settings:* We conduct underwater object detection experiments on the UHTS subset of the RUIE dataset to compare the preprocessing effects of different underwater image enhancement algorithms on high-level visual understanding. The UHTS dataset consists of 300 underwater images, of which 270 images are selected for network training and the left 30 images are used for testing. This article uses the YOLOV3 [60] algorithm as the baseline to train the underwater object detection network. The entire network is trained on an Nvidia GTX 3080Ti GPU and Intel¹ Core² i7-10700KF CPU@3.80 GHZ 32GB RAM using the PyTorch framework. The detection accuracy is evaluated in terms of mean Average Precision (mAP).

2) *Qualitative Results:* Fig. 7 shows a visual comparison of the object detection results in the scenes enhanced by different algorithms. It can be seen from the comparison that the detection algorithm is prone to missed detection when detecting in the scene enhanced by the algorithms proposed by Li et al. [45], Naik et al. [47], Wang et al. [53], Liu et al. [54], Saleh et al. [46], and Rao et al. [55]. The results enhanced by the algorithm proposed in this article are clearer and have high contrast, and the object detection algorithm can well predict the position of the object in the image. Experimental results show that the algorithm proposed in this article is beneficial to high-level visual detection tasks.

3) *Quantitative Results:* Table II shows the average detection precision of different underwater image enhancement

TABLE II

OBJECT DETECTION RESULTS ON UHTS ENHANCED BY DIFFERENT ALGORITHMS

Methods	Urchin (AP_{50})	Holothurian (AP_{50})	Mean
Original images	95.57	56.36	75.96
Li et al. [45]	91.95	74.69	83.32
Naik et al. [47]	94.64	61.37	78.00
Wang et al. [53]	89.84	65.37	77.61
Liu et al. [54]	90.12	64.76	77.44
Saleh et al. [46]	94.18	64.22	79.20
Rao et al. [55]	92.71	47.21	69.96
Ours	95.74	78.46	87.10

TABLE III

ABLATION STUDY ON THE RUIE DATASET

GACC	LEGW	UIQM	UCIQE	NOFP
		2.009	0.544	2356
✓		4.774	0.739	7793
	✓	3.401	0.754	7815
✓	✓	5.223	0.771	8039

algorithms. It can be seen from Table II that the algorithm proposed in this article achieved the best result, 87.10, which is 3.78 AP higher than the second highest result and 11.14 AP higher than original images.

D. Ablation Study

To analyze the effectiveness of the algorithm proposed in this article more deeply, we conducted an ablation study on the RUIE dataset, and the experimental results are shown in Table III. The algorithm proposed in this article includes the following modules.

- 1) Gridded adaptive channel compensation (GACC).
- 2) Local entropy-constrained gray world (LEGW).

It can be seen from Table III that the full model of the algorithm proposed in this article achieves the best

¹Registered trademark.

²Trademarked.

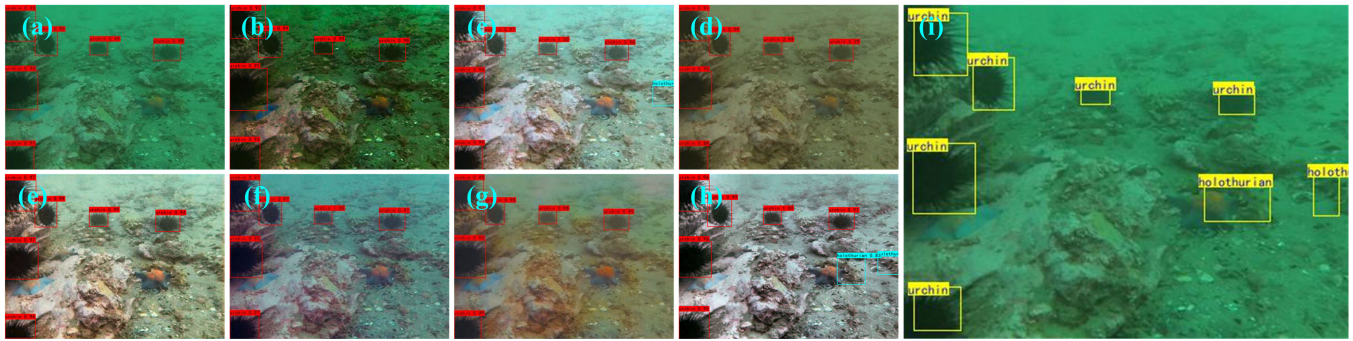


Fig. 7. Object detection results in different enhanced results by different algorithms. (a) Original image. (b) Li et al. [45]. (c) Naik et al. [47]. (d) Wang et al. [53]. (e) Liu et al. [54]. (f) Saleh et al. [46]. (g) Rao et al. [55]. (h) Ours. (i) Ground truth.

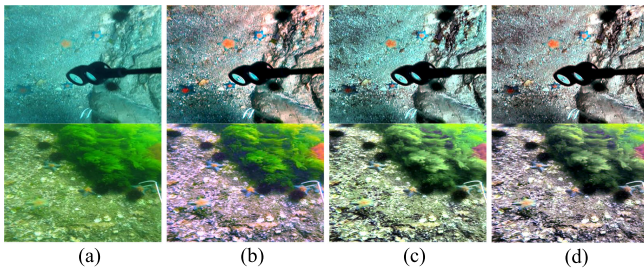


Fig. 8. Qualitative comparison of the multiple modules of the proposed algorithm on the RUIE dataset. (a) Original images. (b) Without GACC. (c) Without LEGW. (d) Full model.

performance on UIQM, UCIQE, and the number of ORB feature points extracted. The qualitative comparison of the contributions of each module is shown in Fig. 8. Combining Table III and Fig. 8, it can be seen that both the gridded adaptive channel compensation algorithm and the local entropy-constrained gray world white balance algorithm proposed in this article can alleviate the color distortion problem.

V. CONCLUSION

This article proposes a multichannel adaptive fusion underwater image enhancement algorithm to solve the color distortion and low contrast caused by underwater light attenuation and scattering. We have conducted experiments on multiple public datasets (UIEB, LSUI, and RUIE) to verify that the image processed by the method proposed in this article has a good improvement in color and contrast, and robustness is better than the comparison methods. At the same time, the method proposed in this article is better than the comparative methods in terms of quantitative evaluation metrics UCIQE, UIQM, and the number of ORB feature points. Although the method proposed in this article is better than the comparative methods in terms of enhancement effect, when conducting real-time experiments on the LSUI dataset, the average time consumed per image is 0.59 s, which cannot meet the real-time requirements. In future research, we will focus on the real-time problem of the algorithm.

REFERENCES

- [1] K. Lebart, C. Smith, E. Trucco, and D. M. Lane, "Automatic indexing of underwater survey video: Algorithm and benchmarking method," *IEEE J. Ocean. Eng.*, vol. 28, no. 4, pp. 673–686, Oct. 2003.
- [2] Y. Y. Schechner and N. Karpel, "Recovery of underwater visibility and structure by polarization analysis," *IEEE J. Ocean. Eng.*, vol. 30, no. 3, pp. 570–587, Jul. 2005.
- [3] X. Fu and X. Cao, "Underwater image enhancement with global-local networks and compressed-histogram equalization," *Signal Process., Image Commun.*, vol. 86, Aug. 2020, Art. no. 115892.
- [4] Z. Liang, Y. Wang, X. Ding, Z. Mi, and X. Fu, "Single underwater image enhancement by attenuation map guided color correction and detail preserved dehazing," *Neurocomputing*, vol. 425, pp. 160–172, Feb. 2021.
- [5] S. Vasamsetti, N. Mittal, B. C. Neelapu, and H. K. Sardana, "Wavelet based perspective on variational enhancement technique for underwater imagery," *Ocean Eng.*, vol. 141, pp. 88–100, Sep. 2017.
- [6] Y. Li, H. Lu, K.-C. Li, H. Kim, and S. Serikawa, "Non-uniform de-scattering and de-blurring of underwater images," *Mobile Netw. Appl.*, vol. 23, no. 2, pp. 352–362, Apr. 2018.
- [7] A. Galdran, "Image dehazing by artificial multiple-exposure image fusion," *Signal Process.*, vol. 149, pp. 135–147, Aug. 2018.
- [8] Y. Huang, F. Yuan, F. Xiao, J. Lu, and E. Cheng, "Underwater image enhancement based on zero-reference deep network," *IEEE J. Ocean. Eng.*, vol. 48, no. 3, pp. 903–924, 2023.
- [9] Y. Dai et al., "Coastal phytoplankton blooms expand and intensify in the 21st century," *Nature*, vol. 615, no. 7951, pp. 280–284, Mar. 2023.
- [10] A. Galdran, D. Pardo, A. Picón, and A. Alvarez-Gila, "Automatic red-channel underwater image restoration," *J. Vis. Commun. Image Represent.*, vol. 26, pp. 132–145, Jan. 2015.
- [11] P. Drews Jr., E. do Nascimento, F. Moraes, S. Botelho, and M. Campos, "Transmission estimation in underwater single images," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops*, Dec. 2013, pp. 825–830.
- [12] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2010.
- [13] Y. Peng, K. Cao, and P. C. Cosman, "Generalization of the dark channel prior for single image restoration," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2856–2868, Jun. 2018.
- [14] C. Dai, M. Lin, X. Wu, Z. Wang, and Z. Guan, "Single underwater image restoration by decomposing curves of attenuating color," *Opt. Laser Technol.*, vol. 123, Mar. 2020, Art. no. 105947.
- [15] K. Ke, C. Zhang, Y. Wang, Y. Zhang, and B. Yao, "Single underwater image restoration based on color correction and optimized transmission map estimation," *Meas. Sci. Technol.*, vol. 34, no. 5, May 2023, Art. no. 055408.
- [16] H. Yu, X. Li, Q. Lou, and L. Yan, "Underwater image enhancement based on color-line model and homomorphic filtering," *Signal, Image Video Process.*, vol. 16, no. 1, pp. 83–91, Feb. 2022.
- [17] X. Ding, Z. Liang, Y. Wang, and X. Fu, "Depth-aware total variation regularization for underwater image dehazing," *Signal Process., Image Commun.*, vol. 98, Oct. 2021, Art. no. 116408.
- [18] H.-Y. Yang, P.-Y. Chen, C.-C. Huang, Y.-Z. Zhuang, and Y.-H. Shiau, "Low complexity underwater image enhancement based on dark channel prior," in *Proc. 2nd Int. Conf. Innov. Bio-Inspired Comput. Appl.*, Dec. 2011, pp. 17–20.
- [19] H. Wen, Y. Tian, T. Huang, and W. Gao, "Single underwater image enhancement with a new optical model," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2013, pp. 753–756.

- [20] L. Chao and M. Wang, "Removal of water scattering," in *Proc. 2nd Int. Conf. Comput. Eng. Technol.*, vol. 2, Apr. 2010, pp. V2-35–V2-39.
- [21] W. Song, Y. Wang, D. Huang, and D. Tjondronegoro, "A rapid scene depth estimation model based on underwater light attenuation prior for underwater image restoration," in *Proc. Pacific Rim Conf. Multimedia*. Cham, Switzerland: Springer, 2018, pp. 678–688.
- [22] C. O. Ancuti, C. Ancuti, C. De Vleeschouwer, and P. Bekaert, "Color balance and fusion for underwater image enhancement," *IEEE Trans. Image Process.*, vol. 27, pp. 379–393, 2017.
- [23] P. Zhuang, C. Li, and J. Wu, "Bayesian retinex underwater image enhancement," *Eng. Appl. Artif. Intell.*, vol. 101, May 2021, Art. no. 104171.
- [24] W. Zhang, Y. Wang, and C. Li, "Underwater image enhancement by attenuated color channel correction and detail preserved contrast enhancement," *IEEE J. Ocean. Eng.*, vol. 47, no. 3, pp. 718–735, Jul. 2022.
- [25] Y. Wang, J. Zhang, Y. Cao, and Z. Wang, "A deep CNN method for underwater image enhancement," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 1382–1386.
- [26] M. Hou, R. Liu, X. Fan, and Z. Luo, "Joint residual learning for underwater image enhancement," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 4043–4047.
- [27] K. Cao, Y.-T. Peng, and P. C. Cosman, "Underwater image restoration using deep networks to estimate background light and scene depth," in *Proc. IEEE Southwest Symp. Image Anal. Interpretation (SSIAI)*, Apr. 2018, pp. 1–4.
- [28] C. Li, S. Anwar, and F. Porikli, "Underwater scene prior inspired deep underwater image and video enhancement," *Pattern Recognit.*, vol. 98, Feb. 2020, Art. no. 107038.
- [29] X. Liu, Z. Gao, and B. M. Chen, "MLFCGAN: Multilevel feature fusion-based conditional GAN for underwater image color correction," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1488–1492, Sep. 2020.
- [30] M. J. Islam, Y. Xia, and J. Sattar, "Fast underwater image enhancement for improved visual perception," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3227–3234, Apr. 2020.
- [31] M. Yang, K. Hu, Y. Du, Z. Wei, Z. Sheng, and J. Hu, "Underwater image enhancement based on conditional generative adversarial network," *Signal Process., Image Commun.*, vol. 81, Feb. 2020, Art. no. 115723.
- [32] Z. Huang, J. Li, Z. Hua, and L. Fan, "Underwater image enhancement via adaptive group attention-based multiscale Cascade transformer," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–18, 2022.
- [33] Y. Wang, H. Liu, and L.-P. Chau, "Single underwater image restoration using attenuation-curve prior," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2017, pp. 1–4.
- [34] Y. Wang, H. Liu, and L.-P. Chau, "Single underwater image restoration using adaptive attenuation-curve prior," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 3, pp. 992–1002, Mar. 2018.
- [35] D. Akkaynak and T. Treibitz, "Sea-thru: A method for removing water from underwater images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1682–1691.
- [36] B. McGlamery, "A computer model for underwater camera systems," *Proc. SPIE*, vol. 208, pp. 221–231, Mar. 1980.
- [37] J. S. Jaffe, "Computer modeling and the design of optimal underwater imaging systems," *IEEE J. Ocean. Eng.*, vol. 15, no. 2, pp. 101–111, Apr. 1990.
- [38] S. Sharma and T. Varma, "Graph signal processing based underwater image enhancement techniques," *Eng. Sci. Technol., Int. J.*, vol. 32, Aug. 2022, Art. no. 101059.
- [39] Z. Tang, J. Li, J. Huang, Z. Wang, and Z. Luo, "Multi-scale convolution underwater image restoration network," *Mach. Vis. Appl.*, vol. 33, no. 6, p. 85, Nov. 2022.
- [40] J. Han et al., "Underwater image restoration via contrastive learning and a real-world dataset," *Remote Sens.*, vol. 14, no. 17, p. 4297, Aug. 2022.
- [41] L. Peng, C. Zhu, and L. Bian, "U-shape transformer for underwater image enhancement," *IEEE Trans. Image Process.*, vol. 32, pp. 3066–3079, 2023.
- [42] R. Liu, X. Fan, M. Zhu, M. Hou, and Z. Luo, "Real-world underwater enhancement: Challenges, benchmarks, and solutions under natural light," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4861–4875, Dec. 2020.
- [43] H. Li, J. Li, and W. Wang, "A fusion adversarial underwater image enhancement network with a public test dataset," 2019, *arXiv:1906.06819*.
- [44] D. Berman, D. Levy, S. Avidan, and T. Treibitz, "Underwater single image color restoration using haze-lines and a new quantitative dataset," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2822–2837, Aug. 2020.
- [45] C. Li et al., "An underwater image enhancement benchmark dataset and beyond," *IEEE Trans. Image Process.*, vol. 29, pp. 4376–4389, 2019.
- [46] A. Saleh, M. Sheaves, D. Jerry, and M. Rahimi Azghadi, "Adaptive uncertainty distribution in deep learning for unsupervised underwater image enhancement," 2022, *arXiv:2212.08983*.
- [47] A. Naik, A. Swarnakar, and K. Mittal, "Shallow-UWnet: Compressed model for underwater image enhancement (student abstract)," in *Proc. 35th AAAI Conf. Artif. Intell. (AAAI)*, vol. 18, 2021, pp. 15853–15854.
- [48] G. Buchsbaum, "A spatial processor model for object colour perception," *J. Franklin Inst.*, vol. 310, no. 1, pp. 1–26, Jul. 1980.
- [49] C. Gao et al., "The von kries chromatic adaptation transform and its generalization," *Chin. Opt. Lett.*, vol. 18, no. 3, 2020, Art. no. 033301.
- [50] W. Cho, S. Seo, J. You, and S. Kang, "Enhancement technique of image contrast using new histogram transformation," *J. Comput. Commun.*, vol. 2, no. 2, pp. 52–56, 2014.
- [51] S. M. Pizer et al., "Adaptive histogram equalization and its variations," *Comput. Vis., Graph., Image Process.*, vol. 39, no. 3, pp. 355–368, Sep. 1987.
- [52] V. Stimper, S. Bauer, R. Ernstorfer, B. Schölkopf, and R. P. Xian, "Multidimensional contrast limited adaptive histogram equalization," *IEEE Access*, vol. 7, pp. 165437–165447, 2019.
- [53] D. Wang, L. Ma, R. Liu, and X. Fan, "Semantic-aware texture-structure feature collaboration for underwater image enhancement," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 4592–4598.
- [54] R. Liu, Z. Jiang, S. Yang, and X. Fan, "Twin adversarial contrastive learning for underwater image enhancement and beyond," *IEEE Trans. Image Process.*, vol. 31, pp. 4922–4936, 2022.
- [55] Y. Rao, W. Liu, K. Li, H. Fan, S. Wang, and J. Dong, "Deep color compensation for generalized underwater image enhancement," *IEEE Trans. Circuits Syst. Video Technol.*, 2023, doi: 10.1109/TCSVT.2023.3305777.
- [56] M. Yang and A. Sowmya, "An underwater color image quality evaluation metric," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 6062–6071, Dec. 2015.
- [57] Z. Hua, L. Qi, M. Guan, H. Su, and Y. Sun, "Colour balance and contrast stretching for sand-dust image enhancement," *IET Image Process.*, vol. 16, no. 14, pp. 3768–3780, Dec. 2022.
- [58] H. Lu, T. Uemura, D. Wang, J. Zhu, Z. Huang, and H. Kim, "Deep-sea organisms tracking using dehazing and deep learning," *Mobile Netw. Appl.*, vol. 25, no. 3, pp. 1008–1015, Jun. 2020.
- [59] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2564–2571.
- [60] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.